

Study Questions: Documenting the Build

Analytical Questions

1. Why is documentation described as the creation of "external generative models"? What is the criterion for sufficient documentation?
2. How does a lab notebook function as an inference log? What specific elements of the Active Inference loop should it record?
3. Explain why documenting the rationale for design decisions is more valuable than documenting only the decisions themselves. What information is lost when rationale is omitted?
4. What are the key practices for version control of physical prototypes? How do they differ from digital version control systems like Git?
5. Why is uncertainty communication important in technical documentation? What risks arise from documentation that presents designs as definitive when they actually carry significant uncertainty?
6. How does the quality of documentation relate to "bus factor" risk in invention projects? Give a concrete example.
7. Explain the difference between documentation that captures a design and documentation that captures a generative model. Which is more valuable and why?
8. What is the role of audience adaptation in technical communication? How should documentation change when the reader's generative model differs from the writer's?
9. Why is the convention "never erase — correct with single line-throughs" important in lab notebooks? What information does an erasure destroy that a line-through preserves?
10. How do Architecture Decision Records (ADRs) formalize the documentation of design decisions? What are the key fields in an ADR?

Applied Questions

1. You are developing a novel bicycle brake system. Write a design decision record for your choice of brake pad material, including alternatives considered, evidence used, uncertainties remaining, and downstream consequences.
2. An inventor has been working on a project for six months with no documentation. They now need to bring on a collaborator. Design a documentation catch-up plan that would bring the project's external generative model up to date as efficiently as possible.
3. You are reviewing another inventor's lab notebook and find an entry that reads: "Tested prototype. Worked great. Moving on to next version." Critique this entry using the standards from this module. What specific information is missing?
4. A hardware startup is preparing to hand off their prototype to a contract manufacturer. What documentation must they provide? How does the "reconstruction test" apply?
5. Your invention uses a proprietary algorithm. Design a documentation strategy that captures enough of the algorithm's logic for a future engineer to understand and modify it, without revealing trade secrets to unauthorized readers.
6. You are writing a patent application for your invention. How does the lab notebook serve as supporting evidence? What documentation practices strengthen a patent claim, and which practices weaken it?
7. An open source hardware project has comprehensive design files but no decision rationale documentation. A contributor wants to modify the power supply circuit but does not know why the original values were chosen. Describe the problem this creates and how ADRs would prevent it.
8. You need to communicate your prototype's test results to three different audiences: a technical co-founder, a potential investor, and a patent examiner. How would you adapt the same test data for each audience's generative model?

9. Your prototype has gone through 15 versions, but you only have photographs of versions 1, 8, and 15. What information has been lost? Design a version control protocol that prevents this loss going forward.
10. Compare Leonardo da Vinci's notebooks with a modern engineering drawing set. What does da Vinci's documentation include that standard engineering drawings do not? Why is this additional content valuable five centuries later?